

**STATE OF CALIFORNIA
CALIFORNIA REGIONAL WATER QUALITY CONTROL BOARD
CENTRAL COAST REGION
81 Higuera Street, Suite 200
San Luis Obispo, California 93401-5427**

**WASTE DISCHARGE REQUIREMENTS ORDER NO. 01-026
WDID No. 3 440801001**

FOR

**CALIFORNIA DEPARTMENT OF FORESTRY
BEN LOMOND YOUTH CONSERVATION CAMP WASTEWATER TREATMENT FACILITY
SANTA CRUZ COUNTY**

The California Regional Water Quality Control Board, Central Coast Region, (Hereinafter called the Regional Board) finds that:

SITE OWNER AND LOCATION

1. On April 3, 2000, Mike Scarbrough, California Department of Forestry, wastewater facility engineer, filed a Report of Waste Discharge. The report was filed on behalf of California Department of Forestry, Ben Lomond Youth Conservation Camp Wastewater Treatment Facility for authorization to continue discharging treated domestic wastewater within the San Lorenzo River sub-basin.
2. California Department of Forestry (hereafter Discharger), 13575 Empire Grade, Santa Cruz, owns and operates a wastewater treatment facility. The facility includes a grinder pump, lift station, a wastewater treatment plant, a shallow leachfield, and two evaporation/percolation ponds. It is located in Santa Cruz County at 13575 Empire Grade, as shown on Attachment "A" of this Order.

PURPOSE OF ORDER

3. The primary objectives of this order are to: 1) Regulate the discharge of treated domestic wastewater; 2) Review and revise discharge limits; and 3) Review and revise a discharge

monitoring program to evaluate the impact to water quality.

SITE/FACILITY DESCRIPTION

Discharge Type

4. The facility is designed to treat and dispose domestic wastewater from the Ben Lomond Youth Conservation Camp.

Design and Current Capacity

5. The facility treatment system consists of a grinder pump lift station, a wastewater treatment plant, a shallow leachfield, and two evaporation/percolation ponds (see Attachment B).
6. Treated ^{four}effluent is alternately discharged among ~~three~~ sections of leachfield. Two evaporation/percolation ponds shall remain available for wastewater disposal if the need arises.
7. The facility is designed to treat an average dry weather flow of 15,000 gallons per day (gpd) and a peak wet weather flow of 21,000 gpd.

Geology

8. The leachfield disposal area is located on

approximately 0.7 acres of sloping (5-15%) topography consisting of sandy loam soils.

Surface and Groundwater

9. The facility is located in the upper portion of the Peavine Creek watershed, a tributary to the San Lorenzo River watershed. Big Creek, a tributary to Scott Creek, is located approximately 2000 feet west of the disposal area. Scott Creek flows directly to the Pacific Ocean.

10. Depth to ground water was measured at a depth of 140 feet and generally migrates in a westerly direction. No ground water data has been collected near this facility.

11. Storm water flows from the vicinity of the leachfield area and into Big Creek and/or San Lorenzo River watershed. Because the flow is less than 1 million gpd, the facility is not required to have a storm water pollution prevention plan (see storm water control requirements in Specification No. 10 of this Order).

MONITORING AND REPORTING PROGRAM (MRP)

12. The MRP requires influent monitoring, effluent monitoring, and system performance monitoring. No ground or surface water monitoring is proposed at this time.

13. Sampling and inspection results shall be reported quarterly by the 1st day of February, May, August, and November.

BASIN PLAN

[Note: Superscripts (^{BP}) are provided throughout the Order to indicate that the requirements were taken from the Basin Plan. Requirements not referenced are based on staff's best professional judgment.]

14. The Water Quality Control Plan, Central Coastal Basin (Basin Plan), was adopted by the Regional Board on November 17, 1989, and approved by the State Water Resources

Control Board on August 16, 1990. The Regional Board approved amendments of the Plan on February 11, 1994 and September 8, 1994. The Basin Plan incorporates statewide plans and policies by reference and contains a strategy for protecting beneficial uses of State waters.^{BP}

15. Present and anticipated beneficial uses of Big Creek that could be affected by the discharge include:^{BP}

- a. Domestic and municipal supply;
- b. Agricultural supply;
- c. Industrial service supply;
- d. Ground water recharge;
- e. Water contact recreation;
- f. Non-contact water recreation;
- g. Wildlife habitat;
- h. Cold fresh-water habitat;
- i. Fish migration;
- j. Fish spawning;
- k. Rare, Threatened, or Endangered Species;
- l. Hydropower Generation; and
- m. Commercial and Sport Fishing.

16. Present and anticipated uses of groundwater in the vicinity of the discharge include:^{BP}

- a. Domestic supply;
- b. Agricultural supply;
- c. Industrial process supply; and
- d. Industrial service supply.

17. The San Lorenzo Wastewater Management Plan (WWMP), adopted by Santa Cruz County Board of Supervisors, was approved by the Regional Board on April 5, 1995 as Resolution No. 95-04. The WWMP includes findings and recommendations resulting from investigation of elevated nitrate levels in surface water and groundwater in the San Lorenzo River watershed. The WWMP recommends the Regional Board require nitrogen control measures in the issuance of new or revised waste discharge requirements. The WWMP's goal is for at least 50 % reduction in nitrogen from onsite disposal systems.^{BP}

18. This order requires a 50 % reduction of nitrogen in effluent, consistent with Resolution No. 95-04.^{BP}

19. The surface water quality objectives specified in the Basin Plan for Boulder Creek (San Lorenzo Hydrologic Sub-Area), are:^{BP}

Analyte	Value	Units
Total Dissolved Solids	150	mg/l
Sodium	20	mg/l
Chloride	10	mg/l
Boron	0.2	mg/l
Sulfate	10	mg/l

20. The range of median ground water quality objectives in the San Lorenzo Sub-Basin, as specified in the Basin Plan, are reported as follows:^{BP}

Analyte	Range	Units
Total Dissolved Solids	100-250	mg/l
Sodium	10-20	mg/l
Chloride	20-30	mg/l
Nitrate (as N)	1-5	mg/l
Sulfate	10-50	mg/l

21. Neither Scott nor Big Creek have specific surface water quality or median ground water quality objectives designated in the Basin Plan.

CEQA

22. These waste discharge requirements are for an existing facility and are exempt from provisions of the California Environmental Quality Act (Public Resources Code, Section 21000, et seq.) in accordance with Section 15301, Chapter 3, Title 14, of the California Code of Regulations.

EXISTING ORDERS AND GENERAL FINDINGS

23. The discharge has been regulated by Waste Discharge Requirements Order No. 87-84, adopted by the Regional Board on June 12, 1987. The discharge from California

Department of Forestry, Ben Lomond Youth Conservation Camp Wastewater Treatment Facility has been regulated by the Regional Board since November 17, 1978.

24. Discharge of waste is a privilege, not a right, and authorization to discharge is conditional upon the discharge complying with provisions of Division 7 of the California Water Code and any more stringent effluent limitations necessary to implement water quality control plans, to protect beneficial uses, and to prevent nuisance. Compliance with this Order should assure this and mitigate any potential adverse changes in water quality due to the discharge.

25. On January 24, 2001, the Regional Board notified the Discharger and interested agencies and persons of its intent to adopt waste discharge requirements for the discharge, and has provided them with a copy of the proposed Order and an opportunity to submit written views and comments.

26. In a public hearing on May 18, 2001, the Regional Board heard and considered all comments pertaining to the discharge and found this Order consistent with the above findings.

IT IS HEREBY ORDERED, pursuant to authority in Section 13263 of the California Water Code, that the California Department of Forestry, its agents, successors, and assigns, may discharge wastewater at Ben Lomond Youth Conservation Camp Wastewater Treatment Facility, providing compliance is maintained with the following:

[Note: Other prohibitions and conditions, definitions, and the method of determining compliance are contained in the attached "Standard Provisions and Reporting Requirements for Waste Discharge Requirements" dated January 1984. Applicable paragraphs are referenced

in Provision No. E.3 of this Order.]

PROHIBITIONS

1. Discharge of wastewater to the treatment system from other than California Department of Forestry, Ben Lomond Youth Conservation Camp Wastewater Treatment Facility, is prohibited.
2. Discharge of wastewater is prohibited outside of the leachfield and evaporation/percolation pond disposal areas shown on Attachment B.
3. Wastewater on the soil surface of the leachfield disposal area is prohibited.
4. Bypass of the collection and treatment systems is prohibited.
5. Discharge of untreated or partially treated wastes directly to the leachfield and/or pond disposal area is prohibited.

SPECIFICATIONS

Effluent Limitations

1. The daily flow rate, averaged over each month, shall not exceed 15,000 gallons per day.
2. Effluent discharged to the leachfield shall not have a pH less than 6.5 or greater than 8.3.^{BP}
3. Effective July 31, 2001, Total Nitrogen in wastewater shall be reduced by at least 50 percent prior to subsurface disposal.^{BP} Compliance will be determined from samples taken at points before and after extended treatment.

Ground Water Limitations

4. The discharge shall not cause a statistically significant increase of mineral constituent (total dissolved solids, sodium, chloride, and nitrate) concentrations in the ground water.
5. The discharge shall not cause ground water to exceed concentrations of chemical constituents limits specified in Title 22, Chapter 15, Article 4, Section 64435, of the

California Code of Regulations as listed below.^{BP}

Constituents	Maximum Contaminant Level (mg/l)
Nitrate (as N)	10

6. The discharge shall not cause concentrations of radionuclides in ground water to exceed limits specified in Title 22, Chapter 15, Articles 5, Section 64443 of the California Code of Regulations.^{BP}
7. The discharge shall not cause concentrations of organic chemicals in ground water to exceed limits specified in Title 22, Chapter 15, Articles 5.5, Section 64444.5 of the California Code of Regulations.^{BP}

System Operation Specifications

8. The wastewater discharge shall be rotated among the three sections of the leachfield daily. *5/3/02 insp.: four fields rotated weekly*
9. High water alarms, readily audible or visible to maintenance personnel, shall be installed at the grinder pump lift station.
10. Standby power or portable pumps shall be available for use when wastewater-pumping facilities fail.

Solids/Solid Waste Control

11. Solids cleaned from treatment unit shall be disposed of at a seepage receiving facility approved by the Regional Board Executive Officer.

Stormwater Control

12. Extraneous surface drainage shall be diverted away from the treatment system and leachfield disposal area.

PROVISIONS

1. Order No. 87-84, "Waste Discharge Requirements for California Department of Forestry and Fire Protection, Ben Lomond Youth Conservation Camp, Santa Cruz County" adopted by the Regional Board on

June 12, 1987, is hereby rescinded.

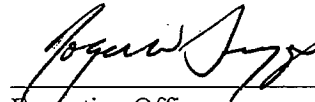
2. The Discharger shall submit an annual Salts Management Program Report. The report shall detail the steps taken to reduce salts in the effluent. The report shall be submitted with the 1st quarter monitoring report for each year.
3. The Discharger shall comply with "Monitoring and Reporting Program No. 01-026", and any amendments thereto, as specified by the Regional Board Executive Officer.
4. The Discharger shall comply with all items of the attached "Standard Provisions and Reporting Requirements for Waste Discharge Requirements" dated January 1984; except Item Nos. A.8 and A.11.
5. Pursuant to Title 23, Division 3, Chapter 9, of the California Code of Regulations, the Discharger must submit a written report to the Regional Board Executive Officer not later than January 1, 2010, addressing:
 - a. Whether there will be changes in the continuity, character, location, or volume of the discharge; and
 - b. Whether, in their opinion, there is any portion of the Order that is incorrect, obsolete, or otherwise in need of revision.

ATTACHMENTS

1. Attachment A, Location Map
2. Attachment B, Site Map
3. Monitoring and Reporting Program No. 01-026
4. Standard Provisions and Reporting Requirements for Waste Discharge Requirements

May 18, 2001

I, **ROGER W. BRIGGS, Executive Officer**, do hereby certify the foregoing is a full, true, and correct copy of an Order adopted by the California Regional Water Quality Control Board, Central Coast Region, on May 18, 2001.



Executive Officer

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Task: 121-01

File: Discharger file; California Department of Forestry / Ben Lomond Youth Conservation Camp

**CALIFORNIA REGIONAL WATER QUALITY CONTROL BOARD
CENTRAL COAST REGION**

**MONITORING AND REPORTING PROGRAM NO. 01-026
FOR**

**CALIFORNIA DEPARTMENT OF FORESTRY
BEN LOMOND YOUTH CONSERVATION CAMP WASTEWATER TREATMENT FACILITY
SANTA CRUZ COUNTY**

INFLUENT MONITORING

Representative influent samples shall be collected before treatment and analyzed for the following:

Parameter	Units	Type of Sample	Minimum Measuring Frequency
BOD, 5-day	mg/l	Grab	Monthly
pH	pH units	"	"
Organic Nitrogen (N)	mg/l	"	* Twice per month
Nitrate (N)	"	"	"
Nitrite (N)	"	"	"
Ammonia (N)	"	"	"
Total dissolved solids	"	"	Quarterly
Total suspended solids	"	"	"

(N) – All nitrogen data shall be reported as nitrogen.

*Staff requires monitoring twice per month for the Nitrogen series (Organic Nitrogen, Nitrate, Nitrite, and Ammonia) for a minimum of 15 months. The samples shall be collected at least 10 days apart. After 15 months, Discharger shall submit a report analyzing the nitrogen removal efficiency of the wastewater treatment plant. Regional Board staff will evaluate the report to determine the need and frequency for continued nitrogen monitoring.

EFFLUENT MONITORING

Representative effluent samples shall be collected prior to leachfield disposal and analyzed for the following:

Parameter	Units	Type of Sample	Minimum Measuring Frequency
Flow to Leachfield	Gallons per day	Metered	Monthly
BOD, 5-day	mg/l	Grab	"
pH	pH units	"	"
Organic Nitrogen (N)	mg/l	"	* Twice per month
Nitrate (N)	"	"	"
Nitrite (N)	"	"	"
Ammonia (N)	"	"	"
Total suspended solids	"	"	Quarterly
Total dissolved solids	"	"	"
Sodium	"	"	"
Chloride	"	"	"

(N) – All nitrogen data shall be reported as nitrogen.

*Staff requires monitoring twice per month for the Nitrogen series (Organic Nitrogen, Nitrate, Nitrite, and Ammonia) for a minimum of 15 months. The samples shall be collected at least 10 days apart. After 15 months, Discharger shall submit a report analyzing the nitrogen removal efficiency of the wastewater treatment plant. Regional Board staff will evaluate the report to determine the need and frequency for continued nitrogen monitoring.

BIOSOLIDS DISPOSAL

The biosolids portion of the wastewater is disposed of at the City of Santa Cruz Wastewater Treatment Facility or the City of Watsonville Wastewater Treatment Facility. Biosolids at these facilities are regulated under Order Nos. 00-044 and 98-019, respectively.

DISPOSAL AREA INSPECTION

Monthly inspections shall be made of the leachfield disposal area for evidence of surfacing effluent. During the inspection, the level of effluent in each leachfield trench shall be recorded. Notes shall be kept of observations and shall be summarized in the Quarterly Monitoring Reports. In the event of a leachfield failure or impending failure, the Regional Board Executive Officer and the County Environmental Health Department shall be notified immediately. In the event of failure, the area shall be posted to prevent public contact with domestic wastewater. Repairs shall be made as quickly as possible. To assure compliance with discharge specifications, the discharger shall describe which leachfield is in use and when the leachfield was last rotated.

REPORTING

Monthly sampling and inspection results shall be reported quarterly by the 1st day of **February, May, August, and November**. Semi-annual sampling results shall be submitted with the February and August quarterly reports. The Annual Salts Management Program Report shall be submitted by the 1st of May.

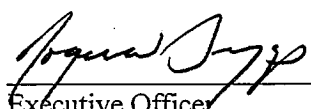
The quarterly reports shall also include a summary of sewage collection or treatment or disposal systems spill(s) during the reporting quarter. At a minimum, the following information shall be reported:

1. Location, volume, date, and time of sewage spill, and the number of times spill has occurred at this location within the last five years.
2. Corrective action(s) taken to stop and mitigate spill and prevent future spills.

If there is no spill during the reporting quarter, a "no sewage spill" statement shall be included in the quarterly report.

All data (weekly, monthly, and semi-annual) shall be submitted to the Regional Board in both hard copy and electronic format. The electronic data submission shall conform to criteria approved by the Central Coast Regional Water Quality Control Board Executive Officer.

ORDERED BY:


Executive OfficerMay 18, 2001

Date